

You need to make revisions to your submission

Your reviewers have provided feedback on your submission. There are revisions you need to make before your submission can progress. We've also emailed thiago.sousa@ifg.edu.br with more details.

We recommend submitting your revisions by 16 Sep 2026.

Feedback from the Editor

Dear Dr Sousa,

Your manuscript, "MOSAIC: single-cell expression from H&E histology", has now been assessed.

We invite you to revise your paper, carefully addressing the comments from the reviewers and the editor. Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained. When your revision is ready, please submit the updated manuscript and a point-by-point response. This will help us move to a swift decision.

Please note that if your manuscript uses any custom or bespoke computational tool or code, or reports a new algorithm, tool, software, or a pipeline (even if individual components are not new), the underlying code must be deposited in a recognised DOI-assigning repository (e.g. zenodo) and linked either from Methods or a dedicated Code Availability section.

Editor Comments

"Some items were found in need of revision by our reviewers. Please elaborate on the source of lineage labels vs evaluation. Provide all needed information to allow full reproducibility, including cell-type annotation procedure, integration parameters and lineage assignment criteria. Uncertainty estimates such as multi-seed evaluation and confidence

intervals may be needed. Please address all comments made by our reviewers in your revision."

-Marina Walther-Antonio
Editorial Board Member
Scientific Reports

Inhouse Editor comments:

1. In-house editorial comment: To aid our readers, and to maximise the accessibility of your manuscript, the title should have a clear, precise scientific meaning and should not contain a colon. Where possible, the title should be read as one concise sentence. Please could you re-write the title ensuring that it is informative and appropriate.

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Anjali Singh
Assistant Editor
Scientific Reports
[Support contact: srep@nature.com](mailto:srep@nature.com)

Submission ID: 296576d6-46b5-4128-b5f7-4500327bef62

Feedback from the Reviewer

Reviewer 1

The manuscript is methodologically sound, appropriately cautious in its claims (noting the modest correlation of 0.205 and the proof-of-concept nature of the work), and now fully compliant with journal policies on data availability and ethics. The data and code availability statements are transparent, and the response to the editorial query has been properly handled. I recommend acceptance without further revision.

Reviewer 2

The manuscript presents MOSAIC, a two-stage framework for predicting a 460-gene expression panel from H&E images at single-cell resolution and subsequently assigning cell lineages from the predicted profiles. The patient/section-disjoint evaluation is a strength, and the authors appropriately acknowledge the relatively modest expression-prediction performance and current limitations. However, several methodological issues need to be addressed before the conclusions can be fully supported.

Major comments

- 1.The Phase B lineage labels were derived from integrated transcriptomic profiles of all seven samples, including the test sample. This may introduce circularity when measured Xenium expression is subsequently used to predict these expression-derived labels. The authors should clarify this issue and, if possible, construct test labels independently.
- 2.The contribution of the spatial-neighbour attention module is not sufficiently demonstrated. Key ablations, such as target-cell only, mean aggregation of neighbours, and attention-based aggregation, should be provided.
- 3.The results are based on a single train/validation/test split and one random seed. Multi-seed experiments, repeated sampling, or additional held-out sections are needed to demonstrate robustness.
- 4.Phase A is mainly evaluated using Pearson correlation. Additional metrics such as Spearman correlation and MAE/RMSE would provide a more comprehensive assessment of expression prediction accuracy.

Minor comments

- 1.Please clarify whether UC6, UC7, and UC1 correspond to completely independent patients or only different tissue sections.
- 2.More details should be provided for the cell-type annotation procedure, including the integration method, Leiden resolution, marker genes, and criteria used for lineage assignment.
- 3.The choice of K=6 neighbours should be better justified, and sensitivity to different K values should preferably be reported.
- 4.Comparisons with previously published methods in Extended Data Table 8 should be interpreted cautiously because the datasets, gene panels, resolutions, and evaluation protocols differ substantially.

Attachment(s):

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Reviewer 3

Modify statement on Lymphatic endothelial cells- both recall and precision are low, not only recall. Add a statement explaining why this is the case, perhaps add limitations of the study section.

Code must be deposited prior to review.

I would suggest to the authors to add a paragraph in the Discussion section, detailing how a better training and validation sets must be designed and collected based on the results of their study.

In my version of the PDF, some tables/references were mixed (ref 1 appearing before Table A4, etc.).